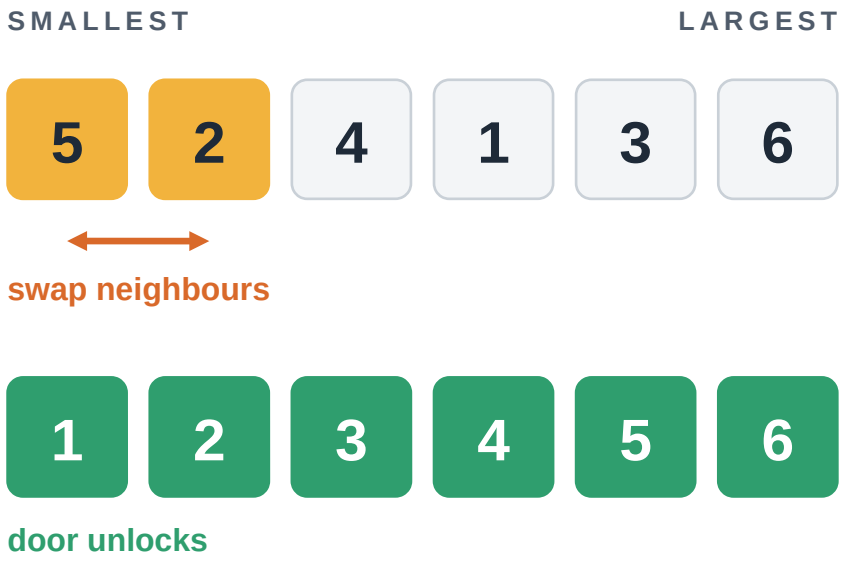
GDIT · GAME PROJECT PRESENTATION

Sort or Stay

A 3D escape-room game where you learn sorting algorithms by sorting real objects — or stay locked in



Group 3 · Course: GDIT — Game project, IIIT Sri City · October 2026

Team member	Roll No.	Contribution (levels)
Premkumar Thota	S20240010243	**Floor 1 · Locked Room — bubble sort (built)** Floor 4 · Control Room — merge sort (next) · player avatars & controls
Vijaya Teja Kameswar	S20240010048	**Floor 2 · Storage Room — selection sort (built)** Floor 5 · Vault — quick sort (planned)
Uday Banu Prakash	S20240010009	**Floor 3 · Library — insertion sort (built)** Debrief cards & scoreboard · pilot study (planned)

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MOTIVATION

Sorting is taught everywhere, but rarely understood

Why this problem

Sorting is the first real algorithm topic in every CS course. Learners memorise the steps, then mix up the algorithms and what each one costs.

Who faces it

First- and second-year Data Structures & Algorithms students, school students starting programming, self-learners and their teachers.

Why it matters

Sorting is where students first meet comparisons, swaps and $O(n^2)$ vs $O(n \log n)$. Gaps here carry into searching, databases and later courses.

Why a game

A game makes the learner do every comparison and swap under the algorithm's rule, with instant feedback, instead of watching an animation.

WHAT THE THREE STUDIES FOUND

76.9% vs 48.7%

reached the top understanding level on Task 1 after sorting cubes by hand in VR, vs. slides and flowcharts (Mukasheva et al., 2023)

4.34 vs 3.72

satisfaction (out of 5) with a gamified sorting module vs. text + GIF animation (Sidorov & Cherkasova, 2026)

Watch ≠ learn

tools that only show an animation leave the learner out of the decision-making (Schöller, 2025)

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PROBLEM STATEMENT

Learners watch sorting but rarely do it themselves

Problem: beginners learn the steps of sorting algorithms but not the rule behind each step or its cost, because today's tools are passive, abstract, or need costly hardware. We need an affordable, hands-on game that checks every move and explains every mistake.

Current limitations

- Animations and bar visualisers are passive: the learner watches, never decides
- 2D card apps check moves but stay abstract and rarely explain why a move is wrong
- VR helps more, but needs a headset for every learner
- Costs (comparisons, swaps) are seldom compared across algorithms

Target users

- 1st–2nd year undergraduates in Data Structures & Algorithms (about 17–21)
- Senior school students starting programming
- Teachers: a class demo, then practice at home

Scope

- In: 5 comparison sorts (bubble, selection, insertion, merge, quick) as 5 floors
- In: single-player 3D desktop game (Windows / Mac) with move checking, hints, debriefs, scoreboard; a small pilot study
- Out: VR/AR, multiplayer, writing code, non-comparison sorts

STATE OF THE ART · PAPER 1 OF 3

Schöller (2025): a card game that checks every move

Bachelor's thesis, Johannes Kepler University Linz (Austria), March 2025 · Ionic + Angular + Capacitor app for phone and web

Problem

- JKU's sorting workshops for kids used a ~20-year-old Java desktop tool: it froze, needed Java + a .jar, desktop only
- Text and pseudocode leave learners passive

Approach

- Face-down cards; at most two open (like two variables)
- Actions: open, swap, pin (current minimum), finalize
- Algorithm mode checks each action against a pre-built action queue; Free Play has no rules

Key contribution

- Step-level checking: mistake +1 and an alert listing the allowed moves
- Live counters: comparisons, swaps, mistakes
- Workshop mode: teacher collects each student's stats

Limitations

- No study of learning with users
- Abstract 2D cards, no story or world
- Feedback says what is allowed, not why
- Only bubble, selection, insertion

CARD STATES



ACTION QUEUE · SELECTION SORT [3, 2, 1, 5, 4]

```
open([0,1])
pin(1)
open([2])
pin(2)
open([3]) · open([4])
open([0]) → swap(0,2)
finalize(0)
```

next expected move; any other move = mistake + alert

Redrawn from Schöller (2025), Figs. 7, 14 and §3.4

Mukasheva et al. (2023): sorting cubes by hand in VR

Frontiers in Education 8:1195200 · Unity + Oculus Quest 2 · 150 undergraduates, Kazakhstan

Problem

- Abstract steps and several actions on unordered data at once make bubble and selection sort hard
- Teaching relies on paper traces, slides, videos

Approach

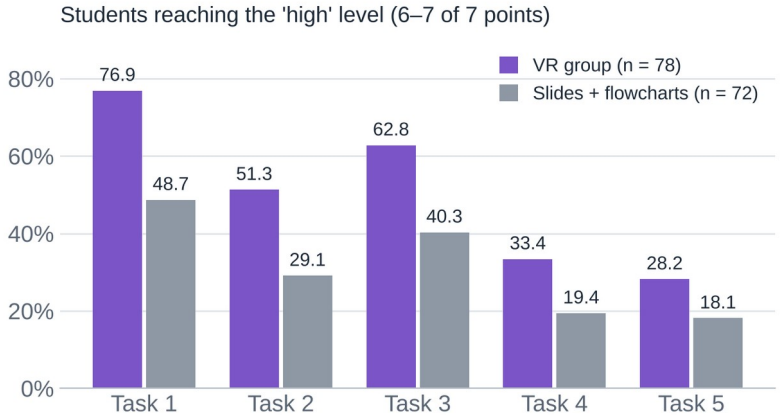
- 'VR sorting': grab numbered cubes with virtual hands and swap them (10-element array)
- Each step shown on a VR board; an error restarts the sort
- 78 VR vs 72 slides + flowcharts; 5 tasks, 7-point scoring

Key contribution

- Every task favoured the VR group (χ^2, $p \le 0.024$)
- Frames VR as 'constructive visualisation': static → dynamic → constructive
- Extended taxonomy: implementation, adaptability, integrity

Limitations

- One university; no delayed test
- Needs a headset per learner
- Only 2 algorithms; no story, score or rewards
- Transfer tasks 4–5 stayed low (≤ 33% high)



+28.2 pts

gap on Task 1 (76.9% vs 48.7%)

α = 0.816

reliability of the 5 tasks (Cronbach)

Mean scores were higher for VR on all five tasks — e.g. Task 2: 5.12 vs 3.35 out of 7.

Chart redrawn from Mukasheva et al. (2023), Table 5

Sidorov & Cherkasova (2026): story levels from theory

Bulletin of Perm University. Math. Mech. Comp. Sci. 1(72): 118–130 · HTML5 / JavaScript web app · pilot with 24 students

Problem

- Learners memorise steps and confuse the logic of different algorithms
- Passive animation is not enough; gamification, AI and learning theory are rarely combined

Approach

- Galperin's stage-by-stage theory: each operation is acted out on objects
- 4 story levels; a mentor character sets each task
- Wrong operations are blocked step by step

Key contribution

- A model that ties a classical learning theory to game mechanics and narrative
- Satisfaction 4.34 vs 3.72 (Mann–Whitney U = 143.5, $p < 0.001$, $r = 0.84$)

Limitations

- Only 24 students; only satisfaction measured
- No test score, no delayed test; medians equal (4.0)
- 2D web, no adaptivity yet

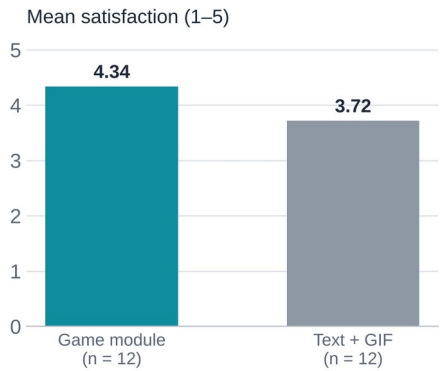


Chart from the paper's group statistics (EG vs CG)

ONE METAPHOR PER ALGORITHM

Bubble
frogs on lily pads

Selection
bridge of magic blocks

Insertion
crystals on an altar

Quick
warriors through gates

Bridge of Selection: pick the smallest unsorted block and place it on the bridge; a wrong pick lowers the bridge's stability.

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SOTA COMPARISON

Where each approach is strong, and where ours fits

	Schöller 2025	Mukasheva et al. 2023	Sidorov & Cherkasova 2026	Sort or Stay (ours)
Technology	Ionic / Angular app, phone + web	Unity VR, Oculus Quest 2	HTML5 / JavaScript web	**Unity 6 3D, desktop (Mac / Win)**
Interaction	Tap cards: open, pin, swap, finalize	Grab and swap cubes with VR hands	Click themed objects	**Realistic avatar; point-and-click, walks to objects**
Game mechanics	Hidden cards, max 2 open; free / algorithm modes	Swap cubes; an error restarts the sort	4 themed levels; bridge 'stability'	**Escape rooms: neighbour-only keys, hidden weights, carry-and-insert books**
Narrative & immersion	None; flat 2D	High immersion, no story	Mentor + fantasy story; 2D	**Locked building, floor by floor; bright 3D rooms**
Feedback	Alert lists allowed moves; counters	Restart on error; steps on a board	Wrong move blocked; theory text	**Says why a move is wrong; hints; debrief vs. optimum**
Adaptivity & AI	None (AI feedback = future)	None	AI used to build it; AI inputs = future	**None yet; adaptive hints planned**
Evaluation	None reported	150 students, 5 tasks, χ^2	24 students, satisfaction only	**Pilot planned: pre / post + delayed test**
Cost to learner	Free (phone / browser)	Headset per learner	Free (browser)	**Free; any laptop**

Takeaway: VR gave the clearest learning gain but costs a headset; the free 2D tools add rules or story but stay flat and give little 'why'. No tool combines all three.

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RESEARCH / DESIGN GAP

Missing: hands-on, explained and affordable at once

	Schöller	Mukasheva	Sidorov	Ours
Learner performs every operation	Yes	Yes	Yes	Yes
3D world without special hardware	No	No	No	Yes
Story that gives the sort a purpose	No	No	Yes	Yes
Algorithm's limits felt (hidden values)	Yes	No	No	Yes
Explains why a move is wrong	Partly	No	Partly	Yes
Cost vs. optimum, across algorithms	Partly	No	No	Yes
Learning measured, incl. delayed test	No	Partly	No	Planned

WHAT THE PAPERS SAY IS STILL MISSING

Schöller 2025, §7

Feedback on how efficient the player's own sorting was, compared with the known algorithms; quick and merge sort as next levels.

Mukasheva et al. 2023, §5

Tasks applying sorting to new problems stayed low; a wider range of tasks is needed. One university only.

Sidorov & Cherkasova 2026

Objective scores and delayed tests with 100+ learners; merge sort; comparing algorithms on the same input.

Gap: a free, hands-on 3D game that makes each rule physical, explains mistakes, compares costs — and is tested on learning, not only liking.

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PROPOSED SOLUTION

Sort or Stay: sort the room, or stay locked in

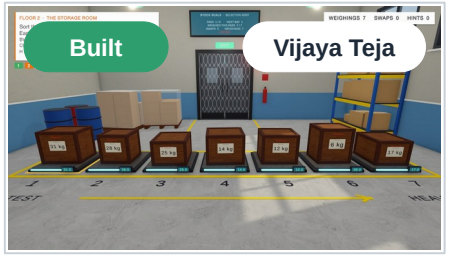
A single-player 3D escape game in Unity 6. You play a realistic human locked in a five-floor building; each floor's exit opens only when its objects are sorted with that floor's algorithm.



Floor 1 · Locked Room

Bubble sort

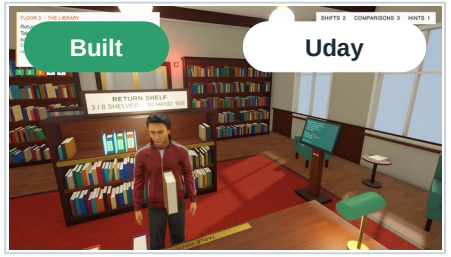
6 keys on hooks; only neighbours may swap



Floor 2 · Storage Room

Selection sort

7 crates with hidden weights; a crane moves the lightest



Floor 3 · Library

Insertion sort

8 books taken in order and slid into shelf gaps

Next

Premkumar

Floor 4 · Control Room

Merge sort

Split, sort halves, merge two queues

Planned

Vijaya Teja

Floor 5 · Vault

Quick sort

Choose a pivot, partition left / right

MAJOR COMPONENTS

Player & controls
realistic avatars, click-to-walk, 1st/3rd person

Rule checker
one puzzle script per algorithm validates each move

Feedback layer
why-messages, hints, gestures, debrief cards

Progress
level manager, game state, cross-floor scoreboard

HOW IT ADDRESSES THE PROBLEM AND GAP

Passive learning → you perform every swap, weigh and insert yourself

Abstract or costly → a 3D world on an ordinary laptop, no headset

No 'why' → each wrong move gets a reason; hints explain the next step

Cost unseen → debrief compares your moves with the fewest possible

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GAME DESIGN

Each room's rule is the algorithm

Mechanics

Pick, swap neighbours, weigh (Q), crane move, fetch (F), insert into a gap, hint (H). Every move is checked.

Dynamics

Players plan passes, re-weigh every crate because selection sort forgets, and watch books shift right.

Aesthetics

Bright, realistic rooms, a realistic human avatar, sound for every thud, scan and page.

Narrative

You wake up locked in. Each floor's door opens only when its objects are in order: sort, or stay.

Interaction

Point and click — the avatar walks over. Number keys, 1st/3rd-person camera, two characters.

Challenges

Only neighbours swap; weights stay hidden until weighed; books come in table order. Wrong moves are slips.

Rewards

Items glow green, the door unlocks, the avatar cheers; a debrief shows your moves vs. the fewest possible.

Progression

6 keys → 7 crates → 8 books → merge → quick: simple $O(n^2)$ first, then divide and conquer.

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GAMEPLAY LOOP

Act, see the result, learn why, face the next pass



EXAMPLE · FLOOR 2, THE STORAGE ROOM (SELECTION SORT)

1 **Player action**
Press Q to weigh every crate left, then click the lightest one (12 kg).

2 **Game response**
Scales show readings; the crane lifts the crate into the next bay; weighings and swaps go up.

3 **Feedback**
A wrong pick: "17 kg is not the lightest crate left" — the avatar scratches his head. H explains the next step.

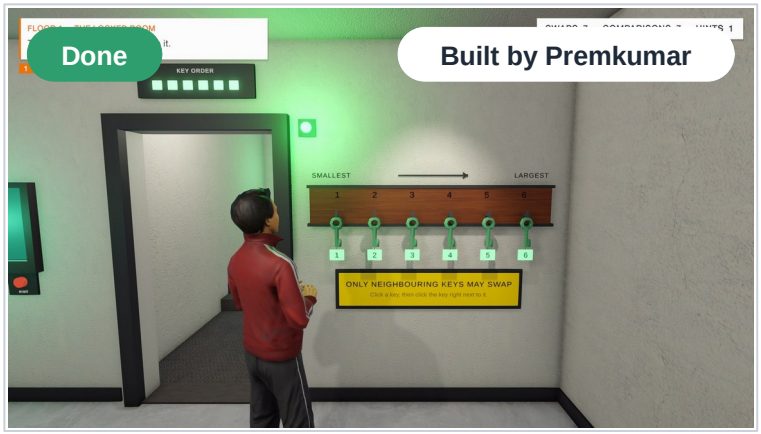
4 **Reward / consequence**
Right: the bay glows green and stays locked in. Wrong: a slip is counted and nothing moves.

5 **New challenge**
Readings reset — the next pass has one crate fewer. After the last pass the doors open to Floor 3.

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CURRENT PROGRESS · COMPLETED

Floors 1–3 are built and playable end to end



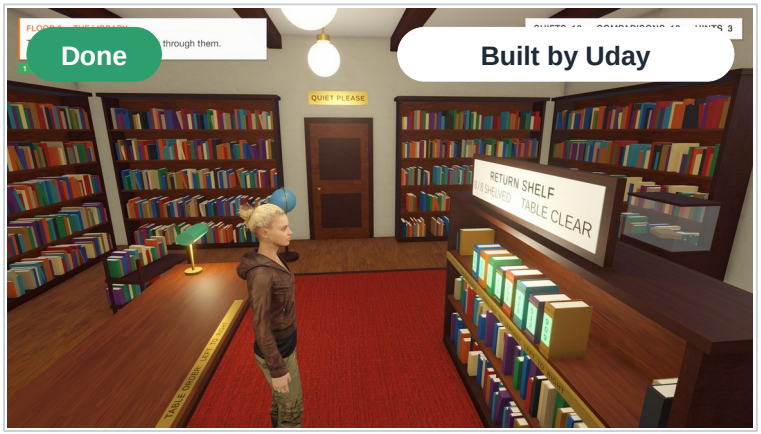
Floor 1 · Locked Room

Bubble sort: all six keys sorted, keys glow green, door open



Floor 2 · Storage Room

Selection sort: bay 1 locked in green; readings reset for pass 2



Floor 3 · Library

Insertion sort: 8 / 8 books shelved, 'table clear'

TEST RUN · 23 OCT 2026 · SCOREBOARD 'FLOORS 1–3 ESCAPED'

7 = 7

Floor 1: swaps = out-of-order pairs, the fewest possible

6 & 21

Floor 2: 6 swaps (at most n − 1) and 21 = n(n−1)/2 comparisons

13 = 13

Floor 3: shifts = out-of-order pairs in the table order

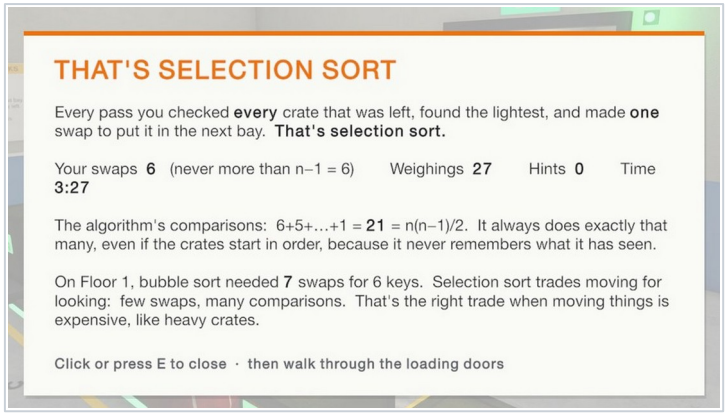
Also done: realistic avatars Alex & Maya · point-and-click with auto-walk · number-key shortcuts · hints · slips · debrief cards · pause menu · editor scene builders

CURRENT PROGRESS · DEMO EVIDENCE

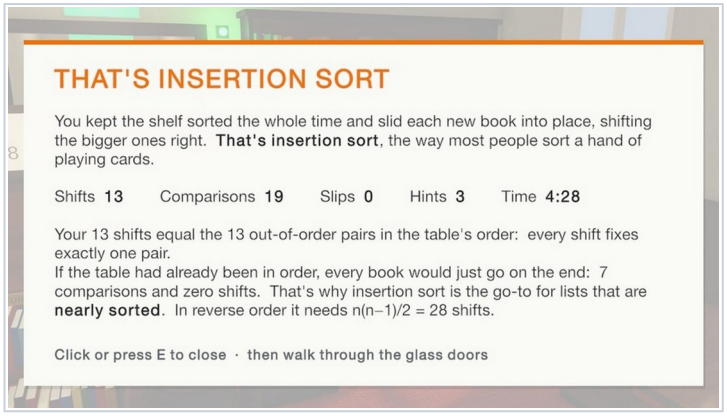
Feedback that explains, not just a score



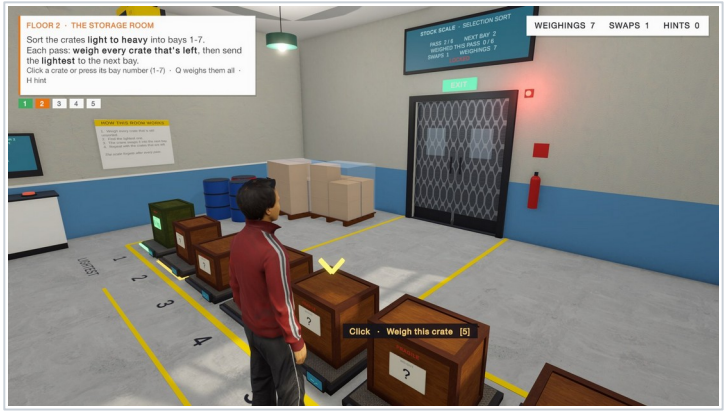
Debrief after Floor 1: "Why 7?" — 7 out-of-order pairs, each swap fixes one



Debrief after Floor 2: 21 = n(n−1)/2 comparisons; compared with Floor 1



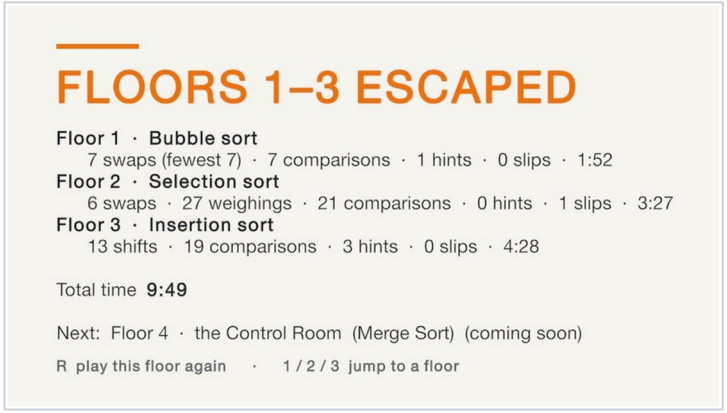
Debrief after Floor 3: shifts = out-of-order pairs; why insertion suits nearly sorted lists



Point and click: hover marker + label, the avatar walks over by himself



Second realistic character (Maya); press C to switch



End scoreboard: all three floors, total time 9:49

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NEXT STEPS

Two floors and a pilot study to go

Work in progress

- Floor 4 · Control Room — merge sort: level design and rule checker (Premkumar)
- Control polish: editor focus click counted as a move; Floor 1 marker overlaps hook numbers (Premkumar)
- Pilot test questions for each floor (Uday)

Remaining work

- Floor 5 · Vault — quick sort, pivot and partition (Vijaya Teja)
- Adaptive hints after repeated slips (Uday)
- Start menu, settings, Windows / Mac builds (all)
- Pilot study: pre / post test, delayed test after a week, questionnaire (Uday leads)
- Final report and demo video (all)

DONE MEANS

- **All 5 floors playable**
- **Pilot study results**
- **Report + demo video**

COMPLETION PLAN (PROPOSED)

Week 1 27 Oct – 2 Nov

Floors 4 and 5: layout + rule checker

Premkumar (F4) · Vijaya Teja (F5) · Uday: tests

Week 2 3 – 9 Nov

Floors 4 and 5 playable; adaptive hints

Premkumar · Vijaya Teja · Uday (hints)

Week 3 10 – 16 Nov

Menus, builds, bug fixes

All three

Week 4 17 – 23 Nov

Pilot study with classmates

Uday leads · all three

Week 5 24 – 30 Nov

Analysis, report, final demo

All three

REFERENCES

Sources

- Schöller, M. (2025). *An Educational Mobile Game about Sorting Algorithms.* Bachelor's thesis, Institute for System Software, Johannes Kepler University Linz, Austria.

- Mukasheva, M., Kalkabayeva, Z., & Pussyrmanov, N. (2023). *Visualization of sorting algorithms in the virtual reality environment.* Frontiers in Education, 8, 1195200. https://doi.org/10.3389/feduc.2023.1195200

- Sidorov, S. V., & Cherkasova, V. A. (2026). *A model for integrating a gamified learning module into the study of sorting algorithms.* Bulletin of Perm University. Mathematics. Mechanics. Computer Science, 1(72), 118–130. https://doi.org/10.17072/1993-0550-2026-1-118-130

ASSETS AND TOOLS

Unity 6 (URP) · Microsoft Rocketbox avatar library (MIT licence) · all screenshots captured from the Sort or Stay Unity project, 23 Oct 2026 · charts redrawn from the papers' reported data